OIKOS

Research article

Increases in interspecific but mixed trends in intraspecific body sizes in German stream macroinvertebrates across two decades

David C. Mittag[1], **Nathan J. Baker**[2], **Laura H. Antão**[3,4], **Lucie Kuczynski**[5,6], **Peter Haase**[1,7], **Ellen A. R. Welti**[1,8]

[1]Department of River Ecology and Conservation, Senckenberg Research Institute and Natural History Museum Frankfurt, Germany
[2]State Scientific Research Institute Nature Research Centre, Vilnius, Lithuania
[3]Organismal and Evolutionary Biology Research Programme, Faculty of Biological and Environmental Sciences, University of Helsinki, Helsinki, Finland
[4]Department of Biology, Faculty of Science, University of Turku, Turku, Finland
[5]Institute of Chemistry and Biology of Marine Environments [ICBM], Carl-Von-Ossietzky University of Oldenburg, Wilhelmshaven, Germany
[6]UMR ENTROPIE, IRD, IFREMER, CNRS, University of La Reunion, University of New Caledonia, Noumea, New Caledonia
[7]Faculty of Biology, University of Duisburg-Essen, Germany
[8]Great Plains Science Program, Conservation Ecology Center, Smithsonian Institution, Bozeman, MT, USA

Correspondence: Nathan J. Baker (nathan93baker@gmail.com)

Oikos
2025: e11596
doi: 10.1002/oik.11596

Subject Editor: Hideyuki Doi
Editor-in-Chief:
Paulo R. Guimaraes
Accepted 23 August 2025

The Anthropocene has accelerated changes both among and within species, altering the functional trait composition of ecological communities globally. Animal body sizes have broad implications for ecosystem structure and functioning, with the temperature–size rule predicting body size declines with warming. We measured individual macroinvertebrate body dimensions from the LTER site 'Rhine-Main-Observatory' between 2001 and 2019 to test for changes in intraspecific biomass for nine common species representing different taxonomic groups. We further assessed changes in interspecific body size of whole macroinvertebrate communities and the relationships between interspecific body sizes and a given species' temporal trend in density across sites. Intraspecific changes in biomass over time were highly variable, with temperature only moderately predicting these trends. Specifically, species with strict univoltine life cycles showed no change over time, while four non-univoltine species experienced the strongest variation in intraspecific body size. In parallel, weighted means of interspecific body sizes of whole macroinvertebrate communities increased over time, with sites with warmer average temperatures being dominated by larger taxa, which also tended to have greater increases in density over time. Our findings do not align with previous predictions of widespread reductions in invertebrate body sizes with rising temperatures, suggesting the temperature–size rule may vary among taxonomic groups, levels of biological hierarchy (intra- versus interspecific), and across sites. Other environmental changes, such as improving water quality, may underlie or counteract shifts in body size, while only two of the sites in our study experienced temperature increases during the study period. Quantifying changes in animal body size within and between species – which is only possible through maintained long-term collections – provides key insights into the processes and dynamics of biodiversity change in a rapidly changing

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world, highlighting the complex dynamics of variation among species and sites, which preclude 'fit-all' predictions of the effects of global change.

Keywords: freshwater; river, global change, insect, temperature, trait

Introduction

Body size is a fundamental characteristic of all animals linked to physiological, life-history and ecological traits (Brown et al. 2004). Shifts in animal body sizes have direct implications for species' populations, ecological communities and ecosystem energy flows (Daufresne et al. 2009, Santini and Isaac 2021). Variation in body size both within and between species can affect nutritional needs, fecundity, dispersal ability, and the likelihood of being consumed by predators (Chown and Gaston 2010, Horne et al. 2017), with implications for ecosystem functioning. For example, larger species or individuals are more effective pollinators (Willmer and Finlayson 2014), while predator–prey interactions and basal food-web services (e.g. decomposition and production) are size-delimited (de Roos and Persson 2002, Scheffer and Carpenter 2003, Woodward et al. 2005). Given the numerous ramifications of animal body size distributions, it is critical to identify the mechanisms that underlie variation in both intra- (i.e. within) and inter- (i.e. between) specific body size through time and space.

Under ongoing climate change, both intra- and inter-specific body size may be shifting with increasing temperatures (Gardner et al. 2011, Martins et al. 2023). The temperature–size rule (TSR) predicts declining body sizes with rising temperatures (Atkinson 1994, Angilletta et al. 2004). This expectation is based on both physiological and macroecological patterns and has primarily been ascribed to smaller animals having increased heat tolerance due to larger surface area to volume ratios. While adherence to the TSR has been documented for fish (Huss et al. 2019) and birds (Weeks et al. 2022), evidence for long-term shifts in invertebrate body sizes remains mixed (Walters and Hassall 2006, Chown and Gaston 2010, Shelomi 2012, Horne et al. 2017, Brehm et al. 2019). Smaller body sizes – with larger surface area to volume ratios – can also be an advantageous trait in freshwater ecosystems due to increased oxygen exchange (Daufresne et al. 2009, Verberk et al. 2021). This need is exacerbated underwater due to the low oxygen availability (>5 orders of magnitude lower than air; Forster et al. 2012) and with increasing metabolic rates as temperature increases (Brown et al. 2004). Oxygen concentrations in water further decrease with increasing temperature, with important repercussions for freshwater communities and ecosystems (Audzijonyte et al. 2018). Due to this additional oxygen constraint, aquatic species are expected to more closely adhere to the TSR than terrestrial species (Forster et al. 2012).

Shifts in animal phenology can further affect body sizes within individual species as well as whole communities by altering the seasonal timing of different life stages, especially for short-lived organisms such as many invertebrates (Chown and Gaston 2010, Horne et al. 2017). Invertebrate life cycles can be adapted to seasonal shifts in resources, as well as hydrological and thermal regimes (Linderholm 2006). Climate change is affecting all these potential cues (Kwon et al. 2019). In particular, warmer temperatures are expected to shift growing seasons earlier, resulting in earlier invertebrate emergence in temperate climates (Linderholm 2006, Verheyen et al. 2018). Shifts in peak biomass of a given population are particularly expected for organisms with univoltine life cycles (i.e. reproducing once a year), for which earlier emergence is expected to lead to increased body size over time for a given sampling period (Chown and Gaston 2010). This would also be expected for semivoltine species (i.e. generations lasting longer than a single year), but with a damped response due to longer generation times (Verheyen et al. 2018). For bi- and multivoltine taxa (i.e. reproducing at least twice a year), predicting body size shifts across years for a given seasonal period in response to earlier emergence becomes more challenging, as sampling can capture different generations (Walters and Hassal 2006). These seasonal shifts can have important repercussions, including on nutrient fluxes and predators relying on invertebrate food sources at particular seasonal periods (Kwon et al. 2019).

Here, we investigated decadal responses in body size by leveraging a long-term collection of freshwater macroinvertebrates from central Germany. We measured body dimensions of 3427 individuals from nine common macroinvertebrate species collected at four stream sites and in two seasons each between 2001 and 2019. As a primary caveat of our intraspecific analysis, we did not record instar stages, thus our body measurements reflect a mixture of both life stages and variation in sizes within a given life stage. While we cannot tease these two mechanisms apart, we sought to identify long-term patterns in intraspecific body size to provide insights into potential ecosystem dynamics. Given heterogeneity in the body plans and phylogenies of different macroinvertebrate groups (e.g. Oligochaeta versus Insecta), we converted macroinvertebrate lengths to biomass estimates using established size–mass relationship equations (with estimated mass hereafter referred to as 'body size'). Additionally, we examined both changes in community weighted means of body size of whole macroinvertebrate communities including 223 taxa (i.e. interspecific trends) and trends in density for taxa occurring in at least 12 sampling years ($n=59$) from the four sites. We quantified both intra- and interspecific trends in body size and variability over time and examined responses to air temperature (Fig. 1). If taxa follow the TSR, intraspecific body size will decrease with increasing temperature (H_{1a}). In

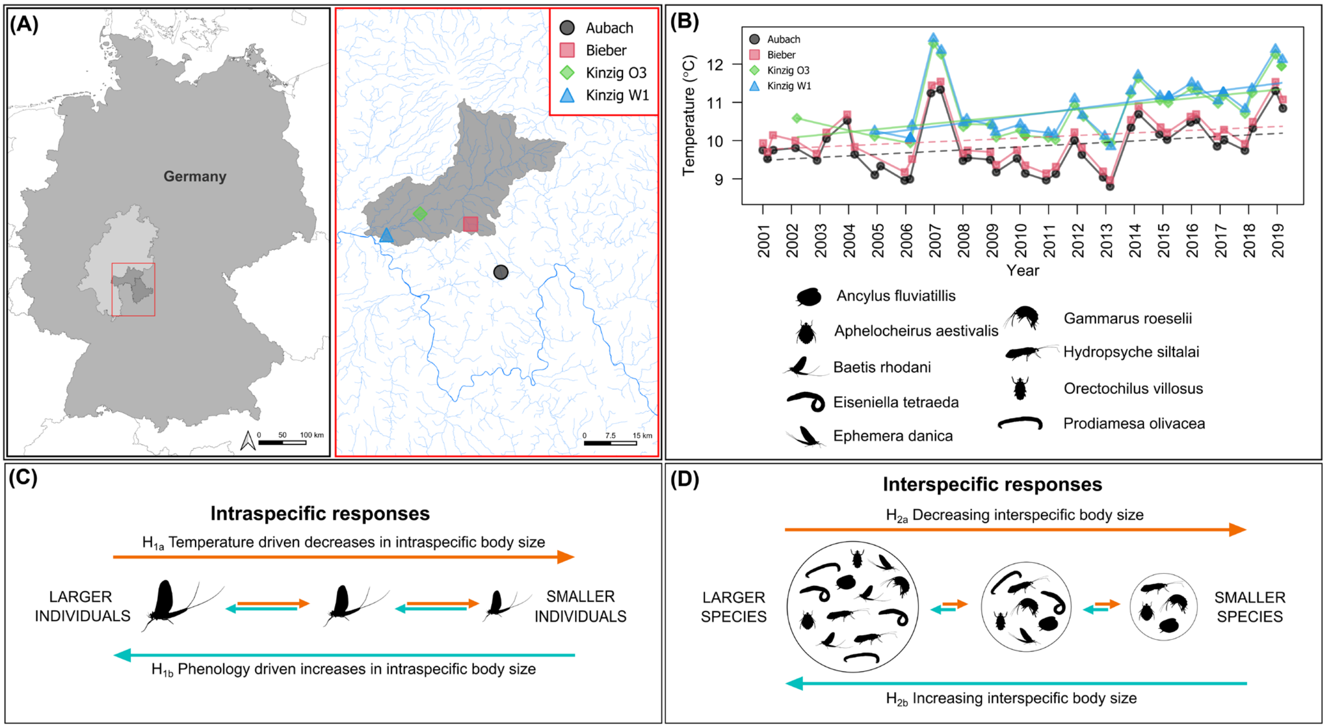

Figure 1. Study location and sampling sites (A), changes in annual temperatures through time (B), hypothesized changes in intraspecific (C) and interspecific (D) invertebrate body size in response to temperature and phenology changes. In (A), the dark gray area within the red box represents the Rhine-Main-Observatory (RMO) area. In (B), solid lines indicate a significant ($p \leq 0.05$) temperature trend. In (C) and (D), invertebrate silhouettes are indicative of the nine species investigated in our intraspecific analysis but do not reflect study results.

contrast, if taxa are emerging earlier in the growing season due to phenological shifts and extended growing seasons, we hypothesized increasing intraspecific body size of strictly univoltine species over time and mixed responses for non-univoltine species (H_{1b}). For interspecific changes, we similarly hypothesized decreases in average community body size, with responses to increasing temperatures resulting in communities dominated by smaller species over time (H_{2a}). Alternatively, increases in interspecific body size over time may occur if other environmental mechanisms underlie community reorganization, such as improvements in water quality following nutrient regulation and restoration actions (Mora-Teddy et al. 2024) (H_{2b}); however this hypothesis is not explicitly tested here due to lack of long-term data on stream chemistry and habitat conditions. Through identifying changes in both species and community patterns of body size, we provide important insight into ongoing changes in ecological functioning.

Material and methods

Study sites

Four sites located in central Germany were chosen as the focal points of our study (Fig. 1). Three sites, Kinzig O3, Kinzig W1 and Bieber, are situated within the Kinzig River drainage basin (~ 1060 km²) and are part of the Rhine-Main-Observatory (RMO), a European Long-Term Ecological Research station (eLTER; Mirtl et al. 2018). The fourth site, Aubach, is located within the 'Spessartwiesen' nature reserve, adjacent to the RMO, and acts as a reference site. All four sites have been continuously monitored over almost two decades, making them some of the longest consecutively observed sites in the region. Both the Aubach (via the Lohr River) and Kinzig rivers are tributaries of the Main River, a major contributor to the Rhine River. This region encompasses diverse environments, ranging from densely populated regions on the outskirts of the Rhine-Main metropolitan area to unimpacted natural landscapes, covering a wide array of land-use types, including urban centers, industrial zones, cultivated farmlands, managed forests and protected areas. The four selected sites reflect this land-use gradient, with Bieber and Aubach representing headwater streams (upstream areas < 50 km²) that have experienced less anthropogenic perturbation, while the two Kinzig sites (upstream areas ranging from 700–900 km²) exhibit higher levels of human impact. Sites ranged in temperatures from the warmest Kinzig W1 (11.9°C ± 5.9 SD in water temperature, averaged over the years with high data availability: 2015–2019), to Kinzig O3 (11.4°C ± 5.5 SD), to Bieber (10.3°C ± 4.5 SD), and to the coolest Aubach (9.9°C ± 3.9 SD). The elevation gradient in the RMO ranges from 100 to 200 m a.s.l., while mean annual air temperature and precipitation are 9.62°C and 837.88 mm, respectively. Further details regarding the RMO can be found at https://deims.org/9f9ba137-342d-4813-ae58-a60911c3abc1.

Field sampling

Macroinvertebrate communities at these sites were sampled biannually (spring: March or April and summer: June or July) between 2001–2019 for Aubach and Bieber (2005 missing for the latter), and between 2005–2019 for the two Kinzig sites. Sampling followed the European Water Framework Directive's standardized multi-habitat sampling protocol, with 20 sub-samples collected from a 100 m section of stream at each site and period (Haase et al. 2004). Specimens were preserved in 70% ethanol, identified primarily to genus and species levels using an operational taxa list (Haase et al. 2006) and deposited in the natural history collections at the Senckenberg research station in Gelnhausen, Germany. Nine focal species were measured for analyses of intraspecific body size, while whole sampled macroinvertebrate communities were examined in our analyses of interspecific body size.

Measuring intraspecific body dimensions

To evaluate intraspecific changes, we measured body dimensions of 3427 individuals from nine freshwater macroinvertebrate taxa. These focal species were selected for their ubiquitous presence across these communities and years, and representativeness of different invertebrate groups (four classes and eight orders) and life histories (Table 1). In total, we measured head size and total body length of 848 *Ephemera danica* (Ephemeroptera), 643 *Baetis rhodani* (Ephemeroptera), 390 *Hydropsyche siltalai* (Trichoptera), 510 *Prodiamesa olivacea* (Diptera), 320 *Aphelocheirus aestivalis* (Hemiptera), and 94 *Orectochilus villosus* (Coleoptera). For three additional species, measured body dimensions varied due to their morphologies. For *Ancylus fluviatilis* (n = 281, Gastropoda), we measured shell length, width, and height; for *Eiseniella tetraeda* (n = 135, Annelida), we measured body length and width; for *Gammarus roeselii* (n = 206, Gammaridae), we measured body length and length of the first antennae. The time series length in years per site varied with species (Supporting information).

When present, 10 individuals per sample were chosen haphazardly (samples were emptied into petri dishes, and the first 10 specimens of the selected species found were measured); when fewer than 10 individuals were in a sample, all were measured. Specimens were measured under a microscope using calipers (0.01 mm accuracy). As *Gammarus roeselii* specimens are usually bent, all specimens were brought into a similar pose and then measured by drawing a line from the specimen's head to the urosome using a microscope camera connected to a computer with the software Cell^A (Electro Optics).

Interspecific body length

To examine changes over time in interspecific body length, we used whole community samples collected from the four stream sites and species level information, namely trait data from the DISPERSE database (Sarremejane et al. 2020) within freshwaterecology.info (Schmidt-Kloiber and Hering 2015). In total, 269 taxa were collected across all years and sites; of these, maximum potential body length data were available for 223 taxa (83%). Data were provided in seven fuzzy coded bins (≤0.25 cm, >0.25–0.5 cm, >0.5–1 cm, >1–2 cm, >2–4 cm, >4–8 cm, >8 cm) to which we assigned single values corresponding to the middle point (respectively: 0.125, 0.375, 0.75, 1.5, 3, 6, 10 cm). For taxa with lengths assigned to multiple bins, we used weighted means to assign a single body length value (e.g. for a taxon in which 33% was assigned to the first bin and 67% was assigned to the second bin, we calculated body length as $0.33 \times 0.125 + 0.67 \times 0.375 = 0.293$ cm).

Length to mass conversions

Given that macroinvertebrate mass has the potential to differ massively between species of identical lengths (or other body dimensions), we converted intra- and interspecific body length to dry mass using published length–mass equations. Across our nine focal species and the 215 other taxa used in our interspecific analyses, length–mass relationships were mostly defined at the family level (63%), followed by order (20%), genus (10%), species (3%), subclass (3%), and other (class, suborder; 1%). Whenever possible, we used length–mass relationships from Benke et al. (1999), but consulted additional literature for taxa not described therein (see the Supporting information for a complete list of sources). Except for Gastropoda, for which dry mass was estimated using different shell dimensions (e.g. shell height or shell width), all other dry mass estimates were derived from body length. Dry mass estimations were obtained via the power function $DM = a \times L^b$, whereby DM was dry mass in milligrams, L was the length of the body dimension in millimeters, and *a* and *b* were respectively the intercept and slope of the linear regression used to estimate the length–mass relationships (Mährlein et al. 2016). Hereafter, we refer to the dry mass estimations converted from body-length dimensions as 'body size'.

Temperature data

Biological datasets of each site were supplemented with monthly aggregated daily mean air temperature data for the period of 2000–2019 extracted from the high-resolution (1 km) CHELSA V2.1 climatic dataset (Karger et al. 2017). Extracted temperature data was used to calculate mean annual temperatures from the 12-months prior to the date of biological sampling. The 12-month interval was selected to cover a uniform period of conditions across an invertebrate's life cycle. For example, if biological sampling was conducted on 12 June of a given year, the mean annual temperature for this day of sampling was calculated using monthly mean temperatures from June of the previous year to May of the sampling year. We compared these annual mean air temperature values with annual means calculated from stream temperature measurements taken hourly or bi-hourly via automatic temperature loggers (HOBO UA-001-64 Pendant loggers), but which were only available from 2011 onwards. In cases where stream temperature time–series contained missing data (due to the loggers malfunctioning or being lost), data were interpolated using the *interpolate_gaps_hourly* function

Table 1. Characteristics and trends of the nine focal species listed from largest to smallest body length. Mean body length and head size are averages ± SE. Trend estimates ± SEs are provided for temporal changes (Δ) in density, body size and head size. Downward and upward arrows indicate significant ($p \leq 0.05$) decreasing and increasing size trends, respectively, while no change is indicated by horizontal arrows. Change in density provides an overall trend for the taxa accounting for sampling seasonality (gls estimate of year for a model of density ~ year + second degree polynomial of Julian date + site). Four species exhibited changes in their densities over years (*Ephemera danica*: $p < 0.001$, *Gammarus roeseli*: $p < 0.001$, *Hydropsyche siltalai*: $p = 0.003$, and *Baetis rhodani*: $p = 0.03$). Detailed model parameters for body length and head size models are provided in the Supporting information.

Species	Group	Mean body size (mg)	Mean body length (mm)	Mean head size (mm)	Δ Density	Δ Body size (mg dry weight)	Δ Body length	Δ Head size	Voltinism	Voltinism references
Orectochilus villosus	Coleoptera	26.6 ± 1.52	10.59 ± 0.27	0.61 ± 0.02	→ −0.04 ± 0.07	→ −1.92 ± 0.07	→ −0.34 ± 0.29	→ 0.03 ± 0.02	univoltine	Tachet et al. 2010
Gammarus roeseli	Amphipoda	15.63 ± 0.8	12.69 ± 0.27	not measured	↗ **0.28 ± 0.08 *****	↘ **−4.2 ± 0.92 *****	↘ **−1.28 ± 0.3 *****	not measured	bivoltine, multivoltine	Pöckl 2002, Grabowski et al. 2007
Ephemera danica	Ephemeroptera	8.16 ± 0.27	14.94 ± 0.20	1.32 ± 0.02	↗ **0.27 ± 0.04 *****	↘ **−0.9 ± 0.37 ***	↘ **−0.64 ± 0.27 ***	↘ **−0.10 ± 0.02 *****	semivoltine	Bennett 2007, Buffagni et al. 2009
Hydropsyche siltalai	Trichoptera	4.69 ± 0.21	9.67 ± 0.17	1.12 ± 0.02	↘ **−0.17 ± 0.06 ****	→ −0.15 ± 0.13	→ −0.04 ± 0.1	→ 0.02 ± 0.01	univoltine	Andersen and Klubnes 1983, Graf et al. 2008
Aphelocheirus aestivalis	Hemiptera	2.27 ± 0.11	6.27 ± 0.14	1.57 ± 0.02	→ 0.06 ± 0.08	→ −0.12 ± 0.14	→ −0.20 ± 0.18	→ −0.02 ± 0.03	univoltine	Tachet et al. 2010
Baetis rhodani	Ephemeroptera	1.14 ± 0.04	5.93 ± 0.08	0.92 ± 0.01	↘ **−0.1 ± 0.04 ***	↘ **−0.14 ± 0.04 *****	↘ **−0.33 ± 0.07 *****	↘ **−0.06 ± 0.01 *****	univoltine, bivoltine, trivoltine, multivoltine	Buffagni et al. 2009, Sand and Brittain 2009
Ancylus fluviatilis	Gastropoda	1.1 ± 0.05	4.8 ± 0.08	not measured	→ 0.1 ± 0.06	→ −0.01 ± 0.05	→ −0.08 ± 0.08	not measured	univoltine	Tachet et al. 2010
Eiseniella tetraedra	Oligochaeta	1.02 ± 0.05	27.29 ± 0.86	2.19 ± 0.06	→ 0.06 ± 0.07	→ −0.09 ± 0.11	→ −1.75 ± 1.96	→ −0.03 ± 0.08	univoltine	Solbé 1971, Tachet et al. 2010, Terhivuo et al. 2022
Prodiamesa olivacea	Diptera	0.81 ± 0.03	9.62 ± 0.14	0.52 ± 0.01	→ −0.01 ± 0.08	↗ **0.15 ± 0.02 *****	↗ **0.99 ± 0.13 *****	↗ **0.04 ± 0.01 *****	univoltine, bivoltine	Schmid 1992, Schmidt-Kloiber and Hering 2015

in the ‘chillR’ R package (Luedeling and Fernandez 2024). Interpolation was only used to provide temperature estimates for gaps smaller than 2% of the time-series. As air and stream temperatures were found to be strongly collinear (Pearson’s $r = 0.748$, $p < 0.001$), and even interpolated temperature data from streams had gaps, we used air temperature data in our analyses. This approach generated linked temperature datasets for all biological sampling events, which were used to explore temporal relationships between intra- and interspecific body size and temperature.

Statistical analyses

We analyzed changes over time in 1) intraspecific body sizes of the nine focal species and 2) interspecific body size structure for the macroinvertebrate communities at each of the four sites. All models were fit in R ver. 4.2.2. (www.r-project.org). For overall intra- and interspecific body size response models, we used generalized linear mixed models with the *lmer* function in the ‘lme4’ package (Bates et al. 2015).

Intraspecifically, we analyzed changes in body size over time for each of the nine focal species using mixed models that included three fixed effects of year, Julian date to account for seasonality, and intraspecific density from the same sample to account for intraspecific density dependence, and a random effect of sampling site to account for repeated sampling. To evaluate intraspecific body size responses to changes in annual temperature, we used a similar model but with year replaced with mean temperature over the 12 months prior to sampling.

Interspecifically, we analyzed overall changes in community weighted means (CWM) of interspecific body size with time, again including year and Julian date as fixed effects and sampling site as a random effect. CWMs of body size were weighted by species’ densities to obtain the body size (dry mass in mg) of an average individual within a sampled community. We modeled CWM of interspecific body size responses to year of sampling for each of the four sites individually using linear regression and including Julian date as a second fixed effect to account for seasonality. In all cases when sampling day of year was included in tests, it was modeled as a second-degree polynomial to account for seasonal trajectories. Finally, we analyzed the density trends over time for species present in at least 12 sampling years per site, representing the 59 most common species and longer time series for more robust trend assessment (White 2019). We then tested whether species density trends were predicted by their body size, with body effects fit as a second degree polynomial to test for increasing abundances of intermediate body sizes, using data from all four sites and including site as a random effect. We used the ‘effects’ package (Fox and Weisberg 2018) to graph body size responses while accounting for site effects.

Results

The nine focal macroinvertebrates ranged in body size from the smallest species, the Diptera *P. olivacea* (average mass: 0.81 mg ± 0.03 SE), to the largest species of the Coleoptera *O. villosus* (26.2 mg ± 1.52 SE). For the 223 taxa for which body size data were available from the trait databases, species size averaged 58.37 mg, with a median of 6.04 mg.

Annual temperature increased at both the larger Kinzig River sites, with air temperatures rising by 1.25°C at Kinzig O3 between 2002 and 2019 (Year estimate = 0.074°C year^{-1}, $p = 0.013$) and by 1.32°C at Kinzig W1 between 2005 and 2019 (Year estimate = 0.094°C year^{-1}, $p < 0.001$). Conversely, air temperatures increased only marginally at the two smaller headwater sites between 2001 and 2019: Aubach (Year estimate = 0.039°C year^{-1}, $p = 0.051$) and Bieber (Year estimate = 0.033°C year^{-1}, $p = 0.106$; Fig. 1b).

Intraspecific body size patterns

There was strong variation in changes in body size over time within species, including declining (three species, all semi- or multivoltine), increasing (one species, a bivoltine taxon), and stable trends (five species, all univoltine; Fig. 2, Table 1). Specifically, and after accounting for the day of year of sampling and intraspecific densities, the two Ephemeroptera, *B. rhodani* and *E. danica*, decreased in both body size (mass), length and head width over time, while the Amphipoda *G. roeselii* decreased in body size and length (head width of *G. roeselii* was not measured; Fig. 2, Supporting information). Five species that were all univoltine did not vary in body size over time in mass or any other measured body dimension (i.e. the Gastropoda *A. fluviatilis*, the Hemiptera *A. aestivalis,* the Oligochaeta, *E. tetraedra*, the Trichoptera *H. siltalai*, and the Coleoptera *O. villosus*), while the Diptera *P. olivacea* increased over time in both body size, length and head length (Fig. 2, Supporting information).

Changes in intraspecific body size over time for these nine species were not well explained by temperature over the previous 12-month period (Supporting information). The Oligochaeta, *E. tetraedra*, the Coleoptera *O. villosus*, and the Amphipoda *G. roeselii* declined in body size and length, while one of the Ephemeroptera, *B. rhodani* also declined in body length and head width during hotter years (Supporting information). However, within individual sites, these species displayed more mixed patterns (Supporting information). For species within individual sites experiencing changes over time in body measurements, *E. danica* at Aubach showed long-term declines in length and *G. roeselii* at Kingzig W1 showed long-term declines in both size and length, while both increased in densities at these sites. In contrast, *H. siltalai* declined in both body size, length, and density at Aubach, providing mixed support for negative effects of density dependence on body size measurements (Fig. 2, Supporting information).

Interspecific body size patterns

Contrary to the mixed temporal patterns in intraspecific body sizes, CWMs of interspecific body size increased over time (year estimate across all four sites = 4.3, SE = 0.73, $p < 0.001$, Fig. 3a) and with air temperature (temperature estimate across all four sites = 2.84, SE = 0.92, $p = 0.002$, Fig. 3b). Intermediate to larger sized species were also the most likely to increase in density over time (Table 1,

Fig. 4). Within individual sites, CWMs increased over time at Aubach (Year estimate = 1.28, SE = 0.42, p = 0.002), Bieber (Year estimate = 6.35, SE = 1.44, p < 0.001), Kinzig W1 (Year estimate = 8.04, SE = 2.42, p < 0.001), and at Kinzig O3 (Year estimate = 3, SE = 1.5, p = 0.046, Fig. 3a), the site with the strongest temperature increases. CWMs increased with temperature within individual sites at Bieber (Year estimate = 5.18, SE = 2.31, p = 0.02) and Kinzig O3 (Year estimate = 2.98, SE = 1.48, p = 0.043), but not at Aubach (Year estimate = 0.05, SE = 0.63, p = 0.94) or Kinzig W1 (Year estimate = 2.81, SE = 2.59, p = 0.28, Fig. 3a). When assessing changes for the 59 most common species that allowed sufficient quantification of temporal trends, we found that density trends were variable among species, but that intermediated sized species (~30–80 mg dry mass) were more likely to increase in their densities over time overall (second degree

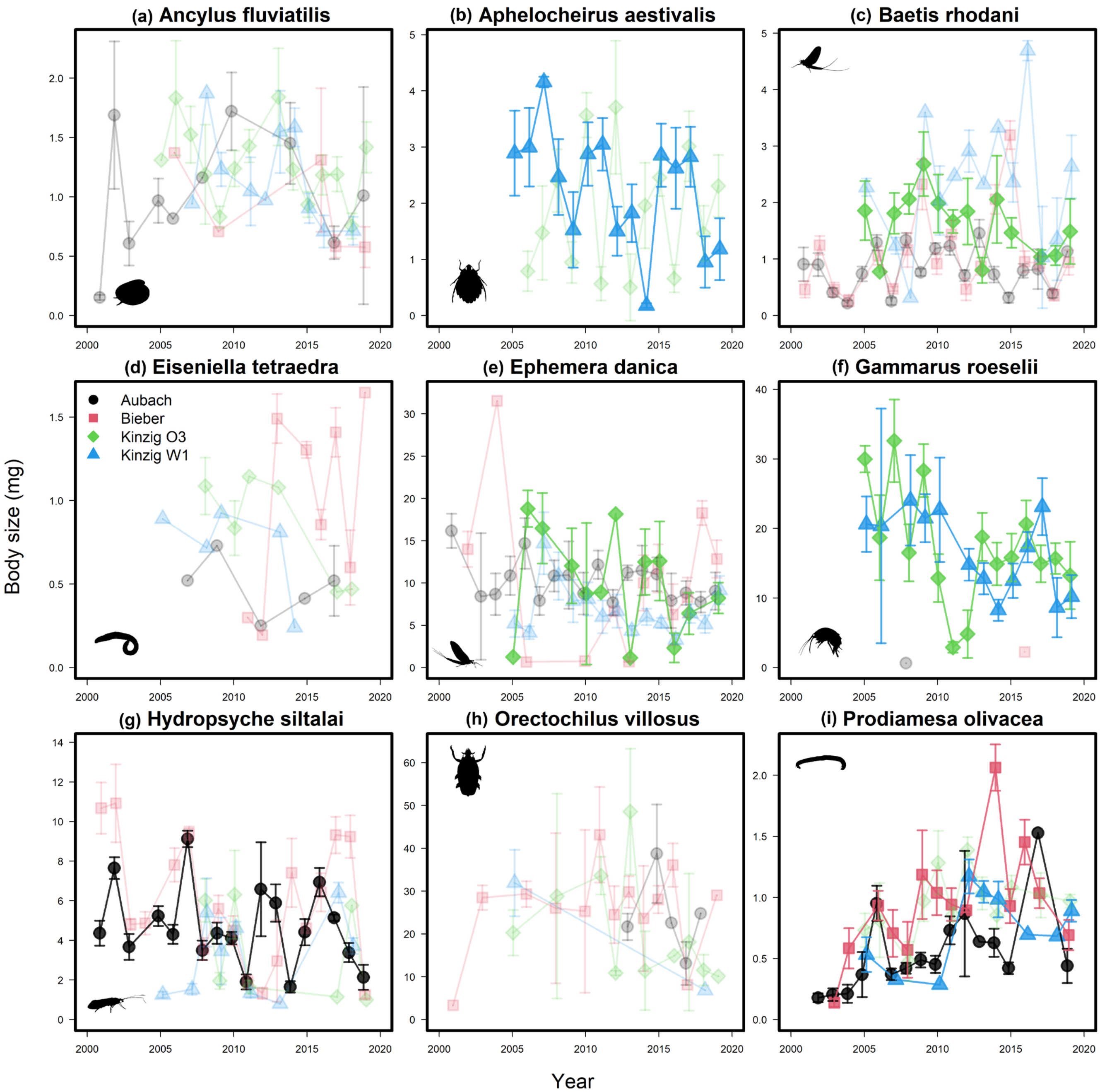

Figure 2. Changes in intraspecific body size (dry mass in mg) over time and in the four study sites for nine freshwater macroinvertebrate species: *Ancylus fluviatilis* (a), *Aphelocheirus aestivalis* (b), *Baetis rhodani* (c), *Eiseniella tetraeda* (d), *Ephemera danica* (e), *Gammarus roeselii* (f), *Hydropsyche siltalai* (g), *Orectochilus villosus* (h), and *Prodiamesa olivacea* (i). Darker points and lines indicate significant ($p < 0.05$) temporal trends.

polynomial of Year estimate = −0.45, SE = 0.17, p = 0.008), but not at the site level (Fig. 4).

Discussion

Intraspecifically, we found divergent trends for nine freshwater macroinvertebrate species measured across two decades, challenging the narrative of universal reductions in body size with warming across spatial scales (Sheridan and Bickford 2011). Instead, we found that species with strictly univoltine life cycles exhibited little change in body size over time, while species with more variable life cycles exhibited greater shifts – both increasing and decreasing in body size. Interspecifically, however, we found temperature-related shifts in community composition resulting from intermediate to larger species being more likely to increase in density over time (i.e. rather than a loss of small species), thus underpinning increases in community-average body size. At the community level, larger macroinvertebrates were more dominant over time and warmer sites (Kingiz O3 and W1) tended to have larger taxa on average. The differences between temporal intra- and interspecific body size variation observed in our study underscores the critical need to identify how and why these changes are occurring, especially given the multitude of ongoing pressures to freshwater communities (Sayer et al. 2025).

Intraspecific variation in body size

Alongside changes to phenology and distribution, body size reductions have been suggested to be a third universal response of organisms to global temperature change (Gardner et al. 2011). We found no unified patterns of temporal changes in intraspecific body size across the nine analyzed species, consistent with some previously reported responses for insects (Martins et al. 2023) and fish (Audzionyte et al. 2020). However, the Oligochaeta *E. tetraedra*, the Coleoptera *O. villosus*, the Amphipoda *G. roeselii* and one of the Ephemeroptera, *B. rhodani*, did decline in body size with temperature, potentially following the temperature–size rule. A lack of overarching responses could be due to the importance of other drivers such as shifts in stream nutrients or runoff (Nguyen et al. 2024).

Many invertebrates have complex life cycles, and co-occurring and often unmeasured changes in potential drivers make it difficult to separate the effect of temperature from other environmental drivers (Daufresne et al. 2009, Horne et al. 2017, Brehm et al. 2018). Interestingly, species that showed temporal changes in intraspecific body size, including both increases and decreases, were associated with non-univoltine life cycles. In contrast, species with univoltine life cycles showed no changes in body size over time, indicating a higher dependence on seasonal predictability. Species with univoltine life cycles may be the most adapted to seasonal cues, whereas species with non-univoltine life cycles may have greater potential for voltinism plasticity (Macgregor et al. 2019). This adds another layer of complexity when attempting to identify clear relationships between temperature or other environmental drivers and intraspecific body size, at least over the 16–19 year periods examined here. The interplay between changes in thermal and phenological regimes can therefore not be easily disentangled.

The three types of response patterns we observed all have multiple plausible causes. First, the five univoltine species (*A fluviatilus, A aestivalis, E. tetraedra, H. siltalai* and *O. villosus*) showed no strong changes in body size through time. This could suggest that univoltine life histories may be driven by fixed seasonal processes such as diel length, with patterns that are not changing over time during these periods. Alternatively, the lack of change could potentially be due to the trends of increasing temperature

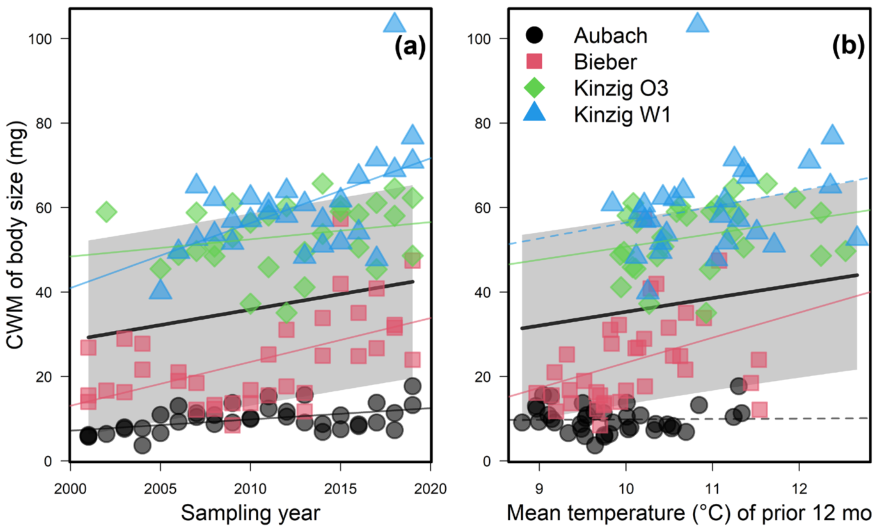

Figure 3. Change in the community weighted mean (CWM) of interspecific body size over time (a) and with water temperature from 2011–2019 (b). Dashed lines indicate non-significant (p > 0.05) changes over time, while solid lines indicate a significant time effect. Thin lines represent site level responses, thick black lines represent overall trends, and the shaded area reflects 95% confidence intervals of overall trends.

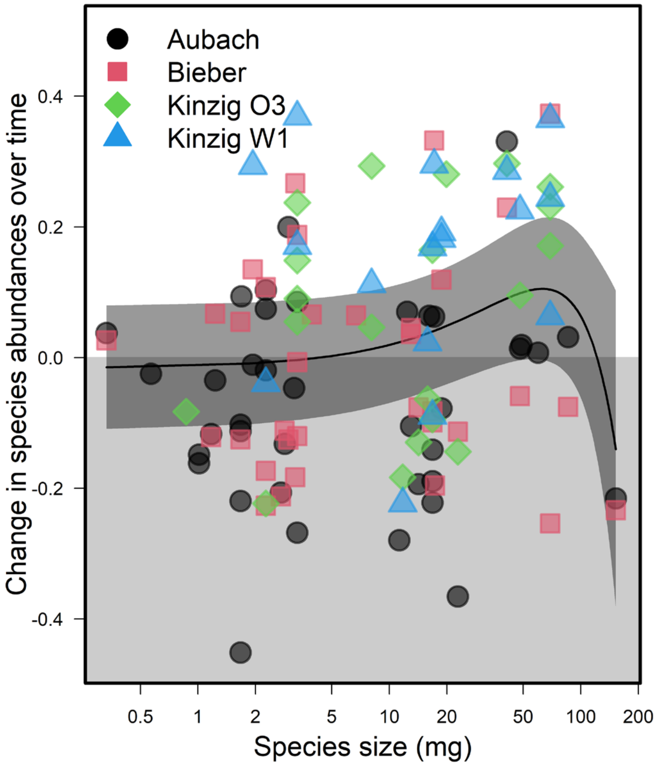

Figure 4. Change in species densities over time (slopes) regressed over species body sizes. Each point represents the estimated slope of one species within one site for species sampled for at least 12 years. The solid black line shows the relationship for all sites.

and shifting phenology balancing each other out. For example, if phenological shifts cause a univoltine species to hatch earlier than usual (Verheyen et al. 2018), then sampled individuals should be larger for a given sampling date. However, if the effects of increasing temperature cause the same individuals to have smaller body sizes (i.e. the TSR; Forster et al. 2012), then the effects of earlier development could be counteracted, leaving univoltine species the same general size, but potentially at different developmental phases. Second, the non-univoltine Diptera *P. olivacea* was the only species that became larger through time. This increase in body size could either result from earlier emergence patterns and longer growing seasons or potentially due to this opportunistic species benefiting from changing environmental conditions. Third, three species (*B. rhodani, E. danica* and *G. roeselii*) which are also non-univoltine, declined in body size through time. This pattern is in accordance with the TSR; however, we did not find evidence of temperature induced shifts in intraspecific body size in our analysis. Alternatively, these species may be emerging earlier in the season in the later years of the time series, with sampling for a given collection period then starting to include individuals collected at the beginning of a new generation cycle.

Interspecific variation in body size

Interspecifically, the CWMs of macroinvertebrate body size increased over the research period at all sites, including the two colder and smaller streams of Aubach and Bieber, as well as at the warmer and larger river sites of Kinzig O3 and W1. Body sizes of macroinvertebrates at the Kinzig sites were also larger overall, but the gap between these and the two colder and smaller streams shrank over the course of study. Larger body sizes may be beneficial for the conditions within the larger rivers of the two Kinzig sites, including sections with stronger and deeper currents. Responses of CWMs of macroinvertebrate community body sizes to air temperature also increased overall and individually within two sites: Bieber and Kinzig O3. Additionally, intermediate to larger species (~30–80 mg dry weight) were the most likely to become more abundant over time, while smaller species and the very largest species were more likely to decline. These results are contrary to the expectation that rising temperatures favor taxa with smaller body sizes (Daufresne et al. 2009) and suggest that the intermediate to larger species in this system are more adapted to warmer water conditions. Alternatively, parallel changes that were untested in this study could include shifts in stream nutrients and flow. Previous work on these streams identified improvements in water quality due to regulation and restoration efforts (Nguyen et al. 2024). The benefits of water quality improvements for oxygen availability and reduced toxicants may have aided recovery of larger stream taxa (Deutsch et al. 2022). These shifts to higher densities of intermediate to larger species may suggest increased top–down pressures on smaller organisms that are prey for macroinvertebrates and may provide more efficiently foraged prey for macroinvertebrate predators, such as fish and birds (Woodward and Hildrew 2002).

Caveats

Despite analysing a unique long-term dataset for freshwater macroinvertebrate body size and abundances, our analysis has several caveats. First, developmental stage was not noted when measuring individuals and we are therefore unable to say with certainty whether the changes in body size are linked to earlier emergence, especially for the species which have multiple generations per year. Regardless, differences in body size responses for univoltine species compared to those species with multiple generations per year may indicate that shifts in phenology are a likely mechanism of intraspecific body size variation through time. This result highlights a key research gap for insects – specifically including developmental phases in biodiversity time series – to tease apart the mechanisms governing changes in body size and shifts in voltinism and phenology. Overall, our results are indicative of some of the known difficulties in teasing out causality in long-term observational data (Nufio et al. 2025). Long-term experiments would be the ideal methodology for identifying the effects of changes in multiple stressors.

Conclusion and future directions

Despite the importance of body size for many physiological and ecological aspects (Woodward et al. 2005), studies of how invertebrate body sizes vary over time have lagged behind work on vertebrate species and communities (Chown and Gaston 2010, Wonglersak et al. 2020). Intraspecifically, increases in environmental temperature can lead to faster developmental rates but smaller size at maturity of ectothermic species (Angilletta et al. 2004). Across freshwater macroinvertebrate taxa collected over 19 years of sampling, we found trends over time in intraspecific body sizes for four non-univoltine species, but not for the five univoltine species, suggesting species with more variable life cycles have outsized effects on community changes. Furthermore, this system saw increases in average body size, and intermediate to larger species were the most likely to increase in density over time. Considering relatively larger taxa may be more likely to be locally lost following degradation, the increases in community level body size we observed may be indicative of local ecological recovery (Staab et al. 2023).

Future work should prioritize collecting additional trait data from museum collections and long-term experiments to identify how and why shifts are occurring in the body size distributions of species and ecological communities. Further, studies assessing both life-stage and body size would be beneficial for teasing apart the mechanisms behind body size shifts, especially considering that changes in environmental conditions are often decoupled from other signals used by animals sensitive to seasonal cues such as diel length. Disentangling the effects of the myriad potential drivers of shifts in animal body size, especially for invertebrates, remains a challenge with many spatial, temporal, and taxonomic gaps. Experimental work, including manipulations of water temperature, light cycles, and nutrient availability can more directly test drivers of invertebrate body size and phenology (Runge et al. 2019). As animal body size has numerous ecological consequences, from an individual's risk of predation to a community's use of resources (Horne et al. 2018), filling these gaps is imperative for forecasting change across global communities.

Acknowledgements – We thank Natalie Kaffenberger and Beatrice Kulawig for their dedication to the long-term sampling of freshwater macroinvertebrates and assistance in the lab.
Funding – Funding was provided to PH and EARW from the EU Horizon 2020 eLTER PLUS (Grant Agreement No. 871128) and from the eLTER PLUS Transnational Access grant: BOSCH: BOdy Size CHanges to EARW, LK and LHA. LK further acknowledge funding by the German Science Foundation funded research unit DynaCom (DFG HI 848/26–2 and HI 848/29–1). LHA acknowledges funding from the Research Council of Finland (grants 340280 and 361416/372215).
Conflict of interest – The authors declare no conflict of interest.

Author contributions

David C. Mittag: Data curation (equal); Writing – original draft (equal); Writing – review and editing (equal). **Nathan J. Baker:** Formal analysis (equal); Visualization (equal); Writing – original draft (equal); Writing – review and editing (equal). **Laura H. Antão:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Writing – review and editing (equal). **Lucie Kuczynski:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Writing – review and editing (equal). **Peter Haase:** Conceptualization (equal); Funding acquisition (equal); Writing – review and editing (equal). **Ellen A. R. Welti:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Visualization (equal); Writing – original draft (equal); Writing – review and editing (equal).

Data availability statement

All data and code to repeat the analyses in this study are included within a GitHub repository (https://github.com/Ewelti/BOSCH) (Mittag et al. 2025).

Supporting information

The Supporting information associated with this article is available with the online version.

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